📘 Day 16: User Behavior Analysis (UBA) & Anomaly Detection

🎯 Objective

Detect suspicious user behavior by identifying deviations from normal activity patterns.

🔍 1. What is UBA?

- **User Behavior Analysis (UBA)** is a cybersecurity technique used to monitor and analyze user activities.
- It helps identify **abnormal behavior** that may indicate:
  - Insider threats
  - Compromised accounts
  - Malicious actions

⚖️ 2. Normal vs Abnormal Behavior

☑️ Normal Behavior

- Login during office hours
- Accessing regular files
- Using known devices/IP addresses

❌ Abnormal Behavior

- Login at unusual times (e.g., midnight)
- Login from unknown or foreign IP
- Accessing sensitive data unexpectedly

🚨 3. Anomaly Detection

- Process of identifying patterns that **deviate from expected behavior**
- Uses logs and activity monitoring to detect:
  - Suspicious logins
  - Unusual data access
  - Sudden behavior changes

🧪 4. Example Scenario

- **User:** employee1

Activity Type	Behavior
Normal	Login at 9 AM (office IP)
Anomaly	Login at 2 AM (foreign IP)
Anomaly	Large data download

👉 This combination strongly indicates suspicious activity.

⚙️ 5. Detection Logic

Basic rule used in SOC systems:

- IF:
  - Unusual login time
  - AND unknown location
- THEN:
  - Trigger alert → **Suspicious behavior**

🛡️ 6. SOC Workflow

◇ L1 Analyst

- Detect anomalies using tools and alerts

◇ L2 Analyst

- Investigate suspicious behavior
- Validate if it is a real threat

◇ L3 Analyst

- Improve detection rules
- Enhance security systems

📊 Activity Analysis Example

Activity	Suspicious?

Login at unusual time	✅ Yes
Large file download	✅ Yes
Quick logout	⚠️ Possibly

👉 Combination of these activities increases risk.

📝 Assignment Answers

1. Explain UBA

UBA is the process of monitoring user behavior to detect abnormal activities that may indicate security threats.

2. Identify Anomalies

- Midnight login
- Login from foreign IP
- Large data download
- Access to sensitive files

3. Suggest Risks

- Account compromise
- Data breach
- Insider threat
- Unauthorized access